WATER CYCLE IN A BAG

An Easy, Fun Science Lesson for Grade 2 (Ages 7–8)

Grade Level	Grade 2 (Ages 7-8)	**Duration**	45 - 60 Minutes
Lesson Type	Hands-On Magic Experiment	**Estimated Cost**	Under ₹20 per student
Context	India-Aligned (Monsoon & Rain Puddles)	**Core Focus**	Simple Observational Science

LESSON OVERVIEW

In this simplified, highly engaging activity, Grade 2 students build a miniature, "magical" water cycle inside a sealed clear plastic bag. Through simple drawing and direct tracking, children witness how warm sunlight turns water into tiny fog droplets and creates actual miniature rain inside their windows. The lesson links beautifully with the Indian monsoon season, helping students discover exactly where the rain outside their windows comes from!

Learning Objectives

By the end of this magical science adventure, students will be able to:

- Identify and explain the three core stages of water: **Evaporation** (going up), **Condensation** (cloud forming), and **Precipitation** (raining down).
- Explain that the warm **Sun** is the main power source that moves the water.
- Set up their own zip-lock bag model and track changes using quick sketches.
- Link the experiment to everyday Indian events like drying clothes or big monsoon downpours.

🌟 The 5 Magic Words for Grade 2:

1. **Evaporation:** Water gets warm and floats up into the sky like invisible steam.
2. **Condensation:** Warm water vapour cools down to make fluffy clouds or tiny water drops.
3. **Precipitation:** Heavy rain or snow falling from the sky.
4. **Sun:** The giant engine in the sky that warms up our water.
5. **Cycle:** A shape like a circle that goes round and round and never stops!

SIMPLE BACKGROUND FOR TEACHERS

For 7-year-olds, keep the water cycle centered around a clear, cyclic narrative: **"Up, Cloud, Down, and Round!"** Water doesn't disappear; it just goes on an endless journey. The exact same water that fills our holy rivers like the Ganga or falls over schools during the monsoon has been recycled for millions of years—even dinosaurs might have splashed in it!

- **Evaporation:** The warm Indian sun heats the oceans and puddles. The water turns into an invisible gas called water vapour and travels up.

- **Condensation:** As it flies high, it gets cold and turns back into tiny visible droplets. These droplets group together to build big rain clouds.
- **Precipitation:** When clouds get too heavy and full, they drop the water back down as splashing monsoon rain!

STEP-BY-STEP LESSON PLAN

Phase 1: The Magic Puddle Hook (5-8 Mins)

1. **Ask the Mystery Question:** *"Imagine you saw a big, muddy rain puddle on the school playground yesterday morning. But by the afternoon, it was completely gone! Where did the water go? Did the ground drink it all, or did it go somewhere else?"*
2. Allow 2 or 3 children to guess. Encourage fun, imaginative answers!
3. **The Action Chant:** Teach the students the 3-step physical water cycle dance to get them moving:

☀️ **EVAPORATION!**
(Raise both hands up high to the sky and wiggle fingers like rising steam)

☁️ **CONDENSATION!**
(Bring hands together over your head to form a fluffy, round cloud)

🌧️ **PRECIPITATION!**
(Wiggle fingers downwards rapidly like heavy falling monsoon rain)

4. Repeat the movement 3 times until the whole classroom is laughing and chanting together.

> 💡 **Teacher Secret:** Grade 2 kids learn incredibly well through physical movement! Use a playful, dramatic voice during the chant to keep energy high.

Phase 2: Fun Classroom Setup (15 Mins)

Give each student or student-pair their experiment pack. Walk them through these simple creation steps:

- 1 **Color the Sky:** Use a black marker to draw a happy, smiling Sun in the top corner, fluffy clouds in the middle, and wavy water waves at the very bottom.
- 2 **Fill the Sea:** Carefully pour exactly half a cup (100 ml) of water into the open bag.
- 3 **Add the Magic Blue:** Drop 1 tiny droplet of blue food coloring into the water. Swirl gently so it looks like a bright blue ocean.
- 4 **Lock It Tight:** Zip the bag completely closed. Press the excess air out gently. *(Teacher check: Double check for leaks!)*
- 5 **Window Magic:** Help students use cello tape to stick their beautiful bags onto a hot, sunny window.

Phase 3: Look & Discover (15 Mins)

1. Let the bags sit in the sun for 25-30 minutes. In India's warm tropical sun (like Tamil Nadu or Mumbai), changes happen fast!
2. Gather the students around the windows and ask these observational questions:
    - *"Look closely! Is the plastic bag getting foggy? Where did that white fog come from?"* (That is **Condensation**!)
    - *"Do you see tiny water drops building up near the clouds at the top?"*
    - *"Watch what happens when the drops get too fat. Are they sliding down like real rain?"* (That is **Precipitation**!)
3. Have students open their notebooks and draw a quick picture of their window bag, using arrows to show the water moving up and down.

> ❓ **Fun Student Discovery:** Advanced kids will notice that while the water at the bottom is blue, the raindrops at the top are completely clear! Explain that when water floats up, it leaves all colors and dirt behind, cleaning itself completely!

Phase 4: Simple Wrap-Up & Homework (10 Mins)

- Review the ultimate rule: **"Water never disappears—it just changes its look!"**
- **The Exit Ticket Game:** Before going to recess or going home, each child must shout out one of our 5 magic words and show its physical hand movement!
- **Home Task:** Leave the bags taped up overnight. Tell the kids to check their windows first thing tomorrow morning and see if it rained inside their bags while they were sleeping!

MATERIALS & EASY RULES

Item Needed	Amount	Fun Grade 2 Notes / Alternatives
Clear Zip-Lock Bag	1 per pair	Medium size, completely clear so they can see the fog clearly.
Tap Water	100 ml (Half Cup)	Plain room temperature water.
Blue Food Color	1-2 Drops	Makes the water look like a real blue ocean! (Optional)
Strong Tape	2-3 Pieces	To stick the bag securely to the sunny window.
Permanent Marker	1 per group	For drawing the happy sun and clouds on the outside.

⚠️ Easy Safety Rules for Little Scientists:

- Blue food dye loves to stain clothes! Keep bags on protective mats or newspaper when pouring.
- If using a table lamp because it is a rainy day, never touch the hot bulb with bare hands. Keep it 20 cm away.
- Ensure the zip-lock is locked tight like a fortress so water doesn't spill onto classroom books.

EASY OBSERVATION LOG (PRINTABLE FOR KIDS)

Teachers can copy this simple table onto the blackboard or print it out for kids to fill with simple emoji faces or drawings:

When?	What does it look like? (Draw it!)	What is happening? (Circle one)
Start (0 Min)	[Draw the flat water level here]	Water is sitting quietly in its blue sea.
Later (30 Min)	[Draw the fog or tiny mist beads]	**Evaporation** (Going Up) / **Condensation** (Fog forming)
End (60 Min)	[Draw the raindrops rolling down]	**Precipitation** (Miniature Window Rain!)